

Data Vortex — Round 1, Phase 1

Exploratory Data Analysis Report

Team/Participant: Sahil Khetal **Dataset:** Social Engine — Users & Posts
(recovered from Archive Node 07)

1. Data Cleaning Summary

The recovered `Social_Engine_Posts_Corrupted.csv` (12,360 rows, 8 columns) contained the following corruption patterns, all resolved in `clean_data.py`:

Issue	Rows Affected	Resolution
Exact duplicate rows	360	Dropped
Missing <code>platform</code>	1,784 (after dedup)	Filled with <code>'Unknown'</code>
Missing <code>text_content</code>	1,688 (after dedup)	Filled with <code>'[No content]'</code>
Missing <code>likes</code>	1,814 (after dedup)	Filled with column median (2,498)
Negative <code>likes</code> values	509 (after dedup)	Corrected via absolute value
Mixed timestamp formats (ISO 8601 / Unix epoch / DD-MM-YYYY)	All 12,360	Standardized to a single datetime format
HTML-escaped text (e.g. <code>&amp; ;</code>)	341	Decoded to plain text

`Social_Engine_Users.csv` (1,500 rows) required no cleaning — no missing values, no duplicate `user_id`s, and every `user_id` referenced in Posts exists in Users (0 orphaned records), confirming referential integrity between the two tables.

Final cleaned dataset: 12,000 posts × 8 columns.

2. Key Insights

2.1 Platform Distribution

Posts are fairly evenly spread across all five recovered platforms (Facebook, YouTube, Twitter, Reddit, Instagram — roughly 2,000–2,074 posts each). The `Unknown` category (platform not recoverable) accounts for 1,784 posts (~14.9% of the cleaned dataset), which should be treated as a data-quality caveat in any platform-specific conclusion.

2.2 Engagement by Platform

Average engagement (likes/shares/comments) is remarkably **consistent across platforms** — no platform dramatically outperforms another:

Platform	Avg Likes	Avg Shares	Avg Comments
Facebook	2,524.1	984.2	506.9
YouTube	2,514.7	1,011.8	504.4
Reddit	2,489.5	1,002.2	511.2
Twitter	2,447.2	1,005.4	506.1
Instagram	2,500.5	1,040.8	499.8

This uniformity suggests engagement in this dataset is not platform-driven — other factors (content type, timing, user base) likely matter more.

2.3 Follower Count vs Engagement — No Correlation

Correlation analysis between a user's `follower_count` and their post's `likes`, `shares`, and `comments` shows **near-zero correlation** across the board (all coefficients between -0.014 and +0.009). This is a notable finding: **having more followers does not predict higher engagement** in this dataset, contrary to typical social media assumptions.

2.4 Geographic Spread

Posting activity is broadly global, with the top locations (Los Angeles, Munich, Shanghai, Barcelona, Houston) each contributing 400+ posts, and no single location dominating — consistent with a globally distributed simulated user base.

2.5 Language Distribution

All 10 languages in the Users table are represented fairly evenly in post volume (1,119–1,369 posts each), with Chinese (`zh`), Japanese (`ja`), and Hindi (`hi`) speakers contributing marginally more posts than English (`en`) speakers.

2.6 Temporal Trend

Post timestamps span **May 2024 to April 2025**, after timestamp standardization. (See `chart_posts_over_time.png` for the full daily trend.)

3. Assumptions Made

1. **Negative likes** were assumed to be a sign-corruption artifact (e.g., a parsing error), not intentionally negative data, since "likes" cannot be negative in any real system — hence `abs()` was applied rather than dropping these rows.
 2. **Missing platform / text_content** were treated as "unrecoverable" fields rather than dropped, since the numeric engagement data (likes, shares, comments) for those rows remained valid and usable.
 3. **Median (not mean)** was used to impute missing `likes` because the distribution is not heavily skewed at the tails (min 0, max 5,000) but median remains the more standard, defensible robust estimator for engagement metrics.
 4. Exact duplicate rows were assumed to be **re-transmission artifacts** from the corrupted intake pipeline, not genuine repeated posts, since every field (including `post_id`) was identical.
-

4. Files Included

- `clean_data.py` — cleaning script with inline justification comments
- `eda.py` — EDA script generating all statistics and charts
- `Social_Engine_Posts_Cleaned.csv` — cleaned posts dataset
- `Social_Engine_Users_Cleaned.csv` — users dataset (unchanged, verified clean)
- `chart_platform_distribution.png`
- `chart_engagement_by_platform.png`
- `chart_posts_over_time.png`
- `EDA_Report.md` — this report